

Miguel Campos

Software Engineer | AI Applications | Full-Stack Development

Python • Generative AI • Local LLMs • React • Node.js • C#

Software Engineer with professional experience developing production web applications and AI-powered software. Experienced in Python, backend systems, Generative AI and full-stack engineering. Designed and developed **Personae**, a privacy-first AI workspace later adopted internally at Kaizen Institute, while contributing to production web platforms, internal engineering tools and AI-assisted development workflows. Passionate about transforming emerging AI technologies into practical products.

EXPERIENCE

Kaizen Institute

Software Engineer

Official title: Web Developer
2025 – Present

- Developed backend functionality and production web features for the company website.
- Improved website performance, Core Web Vitals and technical SEO.
- Built reusable deployment workflows and engineering utilities.
- Introduced AI-assisted software development using Claude Code.
- Evaluated technical solutions and influenced engineering decisions.
- Conceived and developed Personae, later adopted internally to support AI workflows.

EDUCATION

University of Aveiro

Bachelor's in Multimedia and Communication Technologies
2022 – 2025

University of Aveiro

Coursework, Integrated Master's in Electronics and Telecommunications Engineering (not completed)
2019 – 2022

LANGUAGES

PortugueseNative
EnglishFluent

FEATURED PROJECTS

Personae

FEATURED PROJECT

PythonOllamaComfyUISQLiteDockerMCP

Designed and developed a privacy-first AI workspace capable of running entirely offline.

- Local-first architecture
- Persistent conversational memory
- Ollama orchestration
- ComfyUI integration
- Secure local API
- Successfully pitched and adopted internally at Kaizen Institute
- Multiple assistant profiles
- Custom Python HTTP backend
- Translation pipelines
- Local image generation
- Windows deployment automation

DupFree

C#WPFComputer Vision

Desktop application for duplicate and visually similar image detection.

- Perceptual hashing
- GPU accelerated SSIM
- BK-tree indexing
- Fast comparison pipeline

Vehicle Intelligence Data Pipeline

PythonAutomationTranslation

Automated vehicle data collection, translation, normalization and structured dataset generation.

Eco Scan

PHPMySQL

Sustainability platform developed in collaboration with the Municipality of Ílhavo.

SKILLS

AI

Generative AILocal LLMsOllamaPrompt EngineeringMCPHugging FacePyTorch

LANGUAGES

PythonJavaScriptC#PHPJavaC++

FRAMEWORKS

ReactNode.jsExpressFlaskBootstrapAngular

INFRA

DockerDocker ComposeREST APIsGitSQLiteMySQL